

Maximus Jialong Niu

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TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, SQL, C, C++

Frameworks & Libraries: React, Next.js, Tailwind, Storybook, Node.js

Databases: PostgreSQL (pgvector), DynamoDB

Developer Tools: Linux, Git, AWS (S3, IAM, Lambda), Azure (DevOps, Web Apps, AD B2C), Docker, Jest, Supabase

EMPLOYMENT

React Developer

Mar. 2025 – Present

Venuiti Healthcare

Waterloo ON, Canada

- Architected the frontend from scratch for the myMedcan healthcare portal, using **Next.js** and **TypeScript**, integrating a **Salesforce-backed REST API** across **88** endpoints to serve **20,000+** registered users
- Engineered a multi-step appointment booking and payment flow using **XState** and **React Query** to ensure predictable behaviour, integrating **Moneris** for billing, driving **15,000** monthly bookings and **\$168,000** in checkout transactions
- Integrated **Azure AD B2C** authentication on the frontend, eliminating client-side PHI exposure and ensuring **HIPAA** and **PIPEDA** compliance for **20,000+** users
- Built a library of **199+** reusable **React** components and hooks with **Storybook**, shared across multiple developers and pages and meeting client-mandated **WCAG AA** accessibility standards
- Configured an **Azure DevOps CI/CD** pipeline across development, staging, and production environments, automating containerized **Docker** builds and deployments to **Azure Web Apps** with an integrated **Jest** testing suite of **40+** unit tests that run on each deploy

Software Developer

Jan. 2023 – Apr. 2023

Wind River Systems

Ottawa ON, Canada

- Wrote and executed **50+** **C/C++** unit tests on a simulated **x86** board to validate the correctness of **33** Unix utility functions ported to the **VxWorks** RTOS
- Built an internal **Python** and **Flask** automation tool to convert between Markdown formats, processing **500+** pages and streamlining documentation workflows across teams

Full Stack Developer

Sep. 2020 – Dec. 2020

Syntu Software

Waterloo ON, Canada

- Architected a serverless system from scratch for handling sensitive legal evidence using **AWS DynamoDB**, **S3**, **Lambda**, and **IAM**, designed for **PIPEDA** compliance

Software Developer

Jan. 2020 – May. 2020

The Co-operators

Guelph ON, Canada

- Built an automation workflow utilizing **OCR** and **web scraping** to test an internal policy filing application, achieving **400+** policy submissions per hour
- Configured and executed **200+** automated regression tests for the policy filing tool, programmatically validating expected behaviour

PROJECTS

ReWatched – Semantic search over your entire YouTube watch history

- Built an **MV3-compliant Chrome extension** that silently indexes every YouTube video watched by fetching and chunking transcripts, embedding each chunk via **OpenAI**, and storing all vectors locally in **IndexedDB**
- Implemented fully client-side semantic search using **cosine similarity** against the stored index, returning ranked results with match confidence, thumbnails, and deep links that jump to the exact moment in the video

MemoLife – Personal document intelligence via Retrieval-Augmented Generation

- Built a **RAG** pipeline ingesting PDFs, TXT, and MD files using **OpenAI** embeddings and **pgvector** for semantic search, implementing a recursive chunking algorithm with configurable overlap for accurate context retrieval
- Enforced strict per-user document isolation via **Supabase** RLS policies, ensuring users can only query their own uploaded data

EDUCATION

University of Waterloo

Apr. 2024

Bachelor of Applied Science, Honours Computer Engineering

Relevant Courses: Software Architecture, Database Systems, Machine Learning, Systems Programming and Concurrency, Computer Networks, Computer Security, Compilers, Algorithms & Data Structures